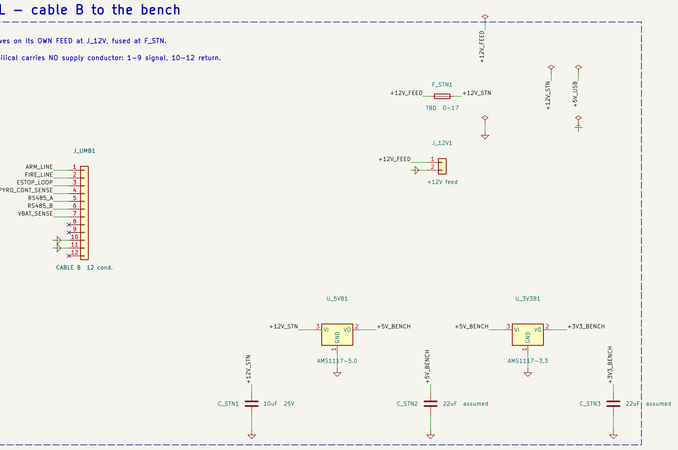
1. UMBILICAL — cable B to the bench

The station supply arrives on its OWN FEED at J_12V, fused at F_STN.

The 12-conductor umbilical carries NO supply conductor: 1–9 signal, 10–12 return.



Bench-side rails: 5 V for the 6N137, 3.3 V for the SP3485.
C_STN1–4 added: this board previously had NO rail decoupling at all.
22 uF values are ASSUMED from the AMS1117 stability note — confirm (O-20).

2. CONTROLS — mechanical, panel-mounted

All four are PANEL controls wired to these terminals, not PCB parts.

S_ARM: removable key, carried by the ignition officer. Closed = bench relay K1 pulls in.



S_FIRE: guarded panel momentary, NO.
Fed FROM ARM_LINE, so FIRE is dead unless ARMED.
SKRPABE010 is NOT acceptable here (D-044).



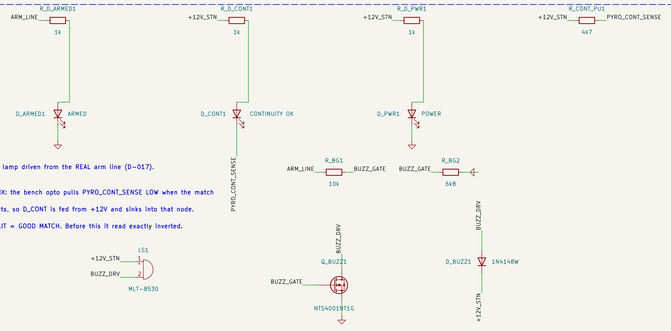
S_ESTOP: latching mushroom, NORMALLY CLOSED.
Opening it drops bench master relay K4.
NC, so a broken wire also stops the bench.

H-2 FIX: the CONTINUITY test button is REMOVED. As drawn it shorted PYRO_CONT_SENSE to ground, which is the lamp-good state, so the test reported a good match whether or not one was connected. A test that cannot fail is worse than no test.

CORRECTION, 9 Nov: the note here previously said sensing is continuous and that current flows through the e-match whenever the bench is powered. THAT IS WRONG and is withdrawn — nothing sources that current. There is NO WORKING CONTINUITY CHECK AT ALL: R_CONT and the opto sit in parallel with the match with no pre-arm source, so the lamp can never light. It fails to the 'no match' reading, the safe direction.

A reviewed pre-arm TEST function is decision D-8, blocked on O-1.

3. INDICATORS & AUDIO



ARMED lamp driven from the REAL arm line (D-017).

H-1 FIX: the bench opto pulls PYRO_CONT_SENSE LOW when the match conducts, so D_CONT is fed from +12V and sinks into that node. LAMP LIT = GOOD MATCH. Before this it read exactly inverted.

Audible ARMED warning — the manual relies on visual cues only.

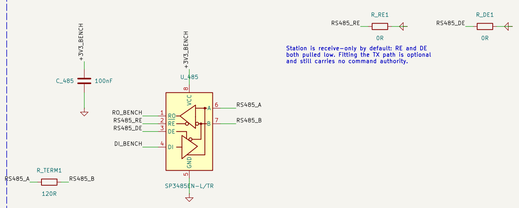
TBD values await O-2 (load cell) and O-6 (isolator part). ARM / FIRE / E-STOP carry no firmware.

4. TELEMETRY — RS-485 in, isolated, out to USB

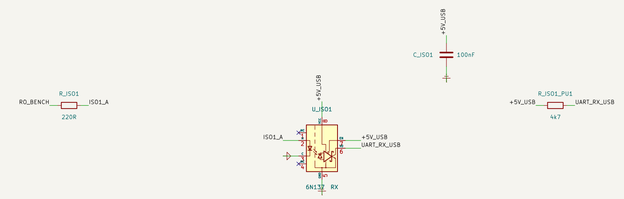
ISOLATION BARRIER down the middle of this block.

LEFT = bench-referenced (GND). RIGHT = laptop side (GNDREF).

No isolated DC-DC: the laptop's USB supplies the isolated side (D-056).



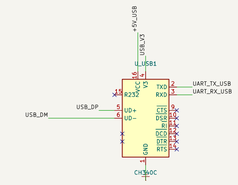
Station is receive-only by default: RE and DE both pulled low. Fitting the TX path is optional and still carries no command authority.

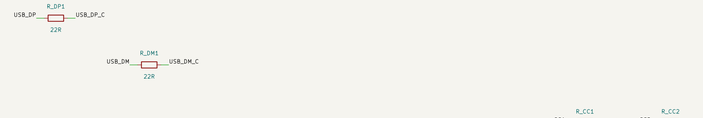


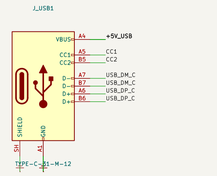
O-21 OPEN, NOT FIXED HERE: this RX path INVERTS. LED conducts on RO HIGH and an enabled 6N137 pulls its output LOW, so an idle mark reaches the CH340C as a continuous break. Telemetry cannot work as drawn. The fix is a part change, not a rewire — see REV-A1-CHANGES.

6N137 is fast enough for 115200 baud. The TLP521 / PC817 already in stock are NOT — do not substitute (O-6).

5. USB-C — laptop side, GNDREF domain









CC pull-downs 5k1 mark this as a UFP sink.
Data series 22R, one ESD diode per line.

6. MEASUREMENT OUTPUT — works with no software at all

PATH 2 IS DEFERRED FOR REV A. The SD log is the measurement.



J_DAQ IS NOT FITTED IN REV A (O-19).
It was wired to umbilical pins 8/9, which reach NOTHING on the bench: no buffer, no source. A connector present at both ends of a dead pair reads as a working analog backup, so it is unpopulated and its pins are flagged rather than left inviting a DAQ.
Pin 3 also exported bench GND to the laptop bench, which bridges the isolation barrier whenever a mains-powered DAQ shares it. Loss of path-2 redundancy for Rev A is recorded in REQUIREMENTS.md.



Battery voltmeter: the pre-test >= 12 V check (manual p. 12), read without walking to the bench.

Review owner: Emilio Guadarrama. Generated by hardware/tools/gen_station.py
TWO GROUND DOMAINS: GND = bench-referenced, GNDREF = laptop/USB. Never joined.
PRELIMINARY — NOT APPROVED. ARM, FIRE and E-STOP are mechanical, panel-mounted.

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Title: TALAVERA v1 — Control Station Board

Size: A1 | Date: 2026-11-10 | Rev: A.1
KiCad 8.0.6 | 1 of 1